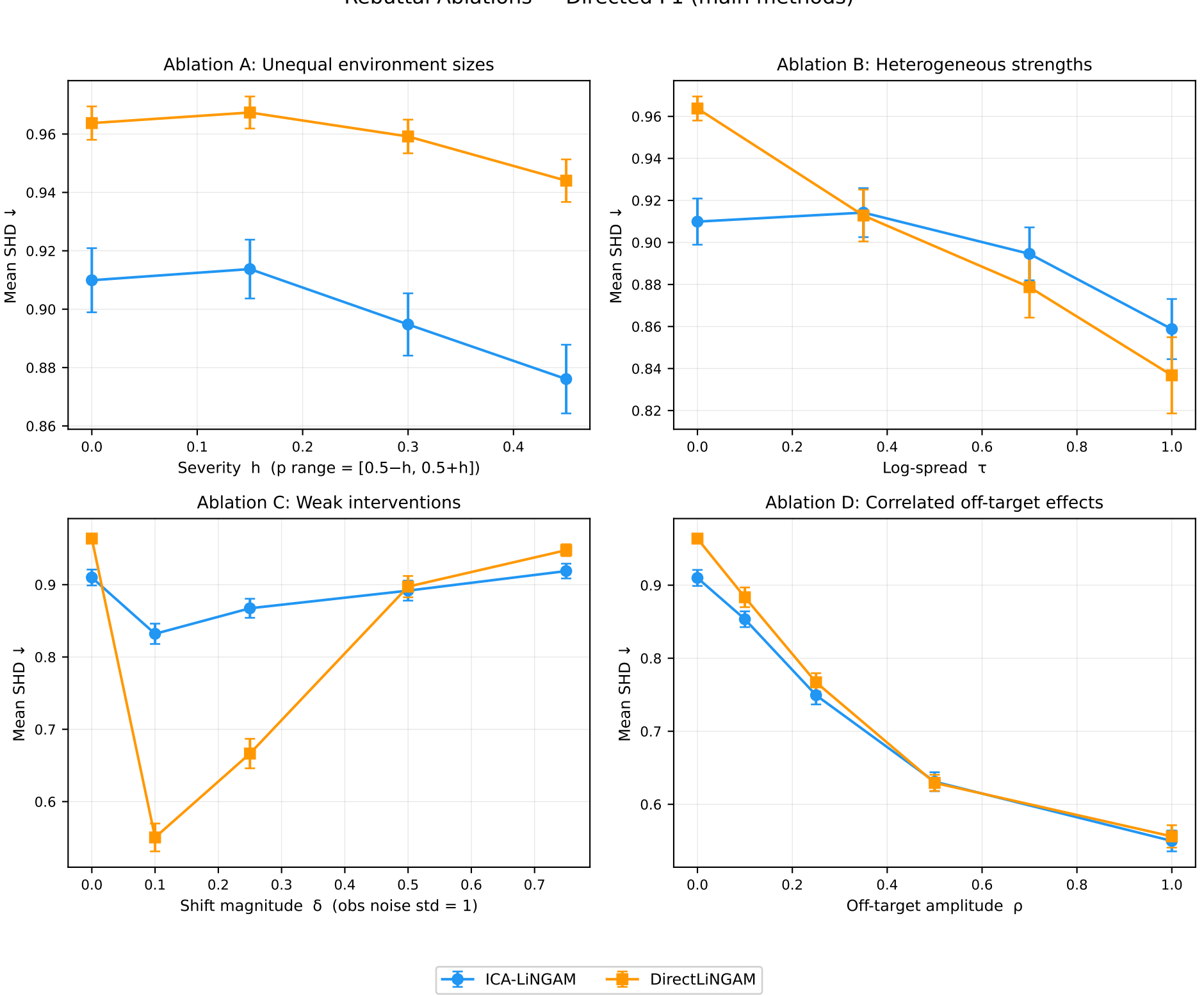
Rebuttal Ablations — Directed F1 (main methods)
Ablation A: Unequal environment sizes
Mean SHD ↓
Severity h (p range = [0.5−h, 0.5+h])
Ablation B: Heterogeneous strengths
Mean SHD ↓
Log-spread τ
Ablation C: Weak interventions
Mean SHD ↓
Shift magnitude δ (obs noise std = 1)
Ablation D: Correlated off-target effects
Mean SHD ↓
Off-target amplitude ρ
ICA-LiNGAM
DirectLiNGAM